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The Moment Closure Hierarchy

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Introduction

Sharkey (2011) demonstrated that epidemic models on networks can be expressed as a hierarchy of moment closure approximations. At each level, the model tracks joint probabilities up to a certain order and closes higher-order moments with an approximation.

Order-1: Individual-based (NIMFA)

The first-order model tracks single-node marginals $\langle S_i \rangle$, $\langle I_i \rangle$ for each node i. The equations for an SIR model on a graph with adjacency matrix A are:

$$\frac{d\langle S_i \rangle}{dt} = -\tau \sum_j A_{ij} \langle S_i I_j \rangle, \qquad \frac{d\langle I_i \rangle}{dt} = \tau \sum_j A_{ij} \langle S_i I_j \rangle - \gamma \langle I_i \rangle.$$

The independence closure replaces the pair moment with a product of marginals:

$$\langle S_i I_j \rangle \approx \langle S_i \rangle \langle I_j \rangle.$$

This is also known as the N-Intertwined Mean Field Approximation (NIMFA). It yields $O(N)$ ODEs.

Order-2: Pair-based (Kirkwood)

The second-order model additionally tracks pair probabilities $\langle S_i I_j \rangle$, $\langle S_i S_j \rangle$ for each directed edge (i, j). Triple moments $\langle A_k B_i C_j \rangle$ that appear in the pair equations are closed with the Kirkwood superposition approximation:

$$\langle A_k B_i C_j \rangle \approx \frac{\langle A_k B_i \rangle \langle B_i C_j \rangle}{\langle B_i \rangle}.$$

This yields $O(N + M)$ ODEs, where M is the number of directed edges. The Kirkwood closure is exact on tree graphs because, in a tree, nodes k and j are conditionally independent given i.

Exact: Gillespie SSA

The exact continuous-time Markov chain can be simulated via the Gillespie stochastic simulation algorithm (SSA). This introduces no approximation beyond sampling noise, which diminishes with repeated runs.

Setup

```julia
using NodeBasedModels
using Graphs
using Plots
using OrdinaryDiffEqDefault
using Random
using Statistics
```

Small graph for visualisation

We use a small graph ($N = 30$, degree 4) so that per-node trajectories are clearly visible and the Gillespie simulation is fast.

```julia
Random.seed!(42)
g = random_regular_graph(30, 4)
net = GraphNetwork(g)

println("Nodes: ", nv(g), "  Edges: ", ne(g))
```

```
Nodes: 30  Edges: 60
```

Per-node trajectories

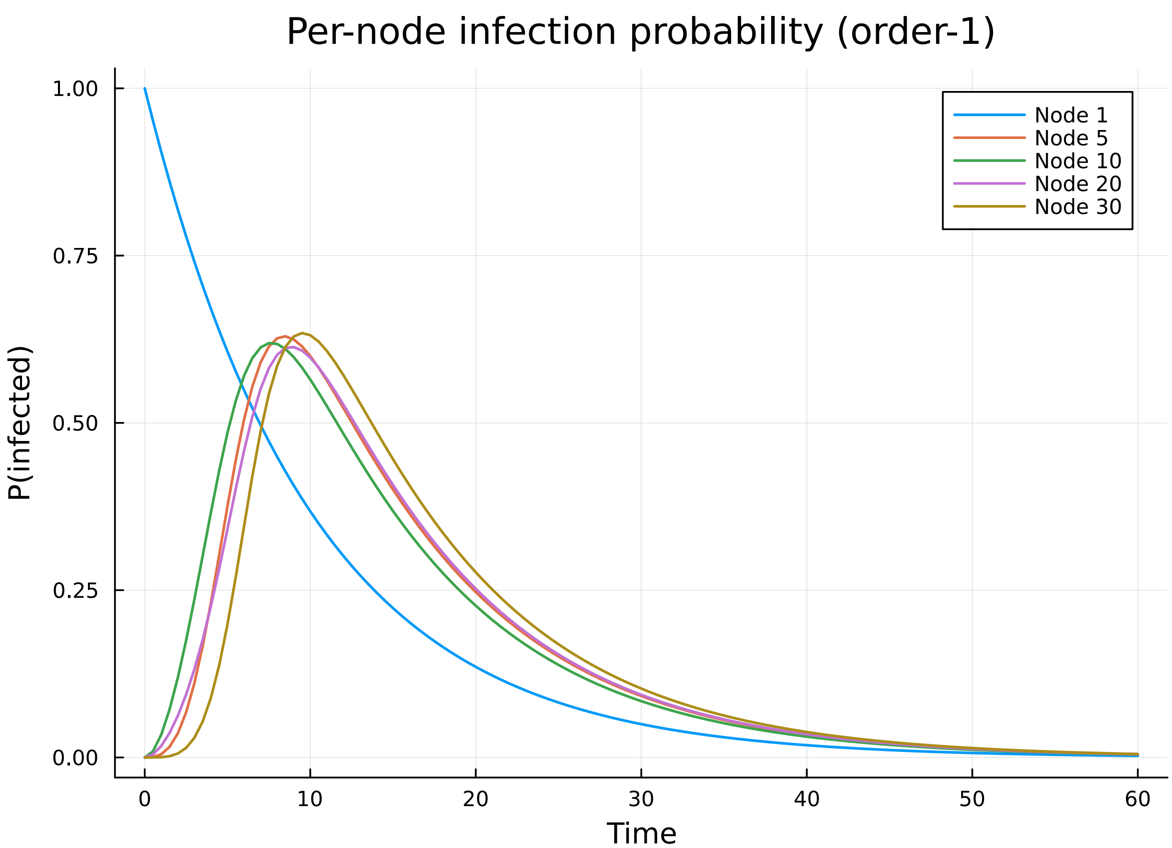
Unlike population-level models, node-based models resolve the epidemic at each node separately. Even on a regular graph (identical degree), different nodes experience different epidemic curves because of their distinct positions in the graph.

```julia
ib_result = generate_individual_based(
    sir_model(), net;
    infection_rate = 0.2,
    recovery_rate  = 0.1,
    initial_infected = [1],
    tspan  = (0.0, 60.0),
    saveat = 0.5
)

t_vals = range(0.0, 60.0, length = length(aggregate(ib_result, :S)))
n_t = length(t_vals)

selected_nodes = [1, 5, 10, 20, 30]
```

```julia
p = plot(xlabel = "Time", ylabel = "P(infected)",
         title = "Per-node infection probability (order-1)")
for node in selected_nodes
    I_node = [node_state(ib_result, node, :I, t) for t in 1:n_t]
    plot!(p, t_vals, I_node, label = "Node $node", lw = 1.5)
end
p
```



Node 1 (the initially infected node) starts with infection probability close to 1 and recovers quickly. More distant nodes have delayed, lower-amplitude curves. This spatial heterogeneity is absent from any population-level model.

Anomalous terms and the independence closure error

The order-1 model replaces $\langle S_i I_j \rangle$ with $\langle S_i \rangle \langle I_j \rangle$. The error in this approximation is the covariance:

$$\mathrm{Cov}(S_i, I_j) = \langle S_i I_j \rangle - \langle S_i \rangle \langle I_j \rangle.$$

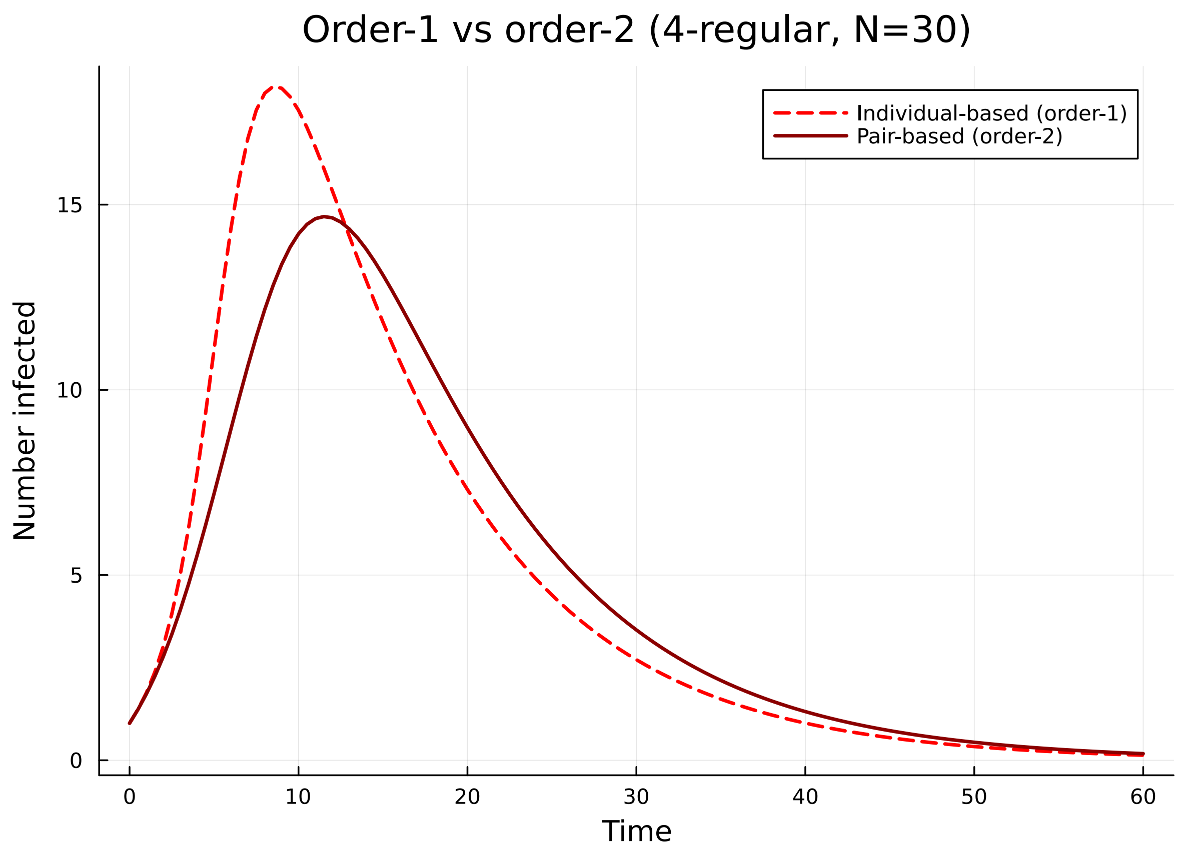
During an epidemic, this covariance is generally positive for neighbours: if i infected j, then j is still likely infected while i is also still infected (a 2-cycle correlation). The independence closure ignores this, so it overestimates the force of infection.

The error grows with the density of short cycles in the graph. We demonstrate this by comparing the individual-based and pair-based models:

```
pb_result = generate_pair_based(
    sir_model(), net;
    infection_rate = 0.2,
    recovery_rate  = 0.1,
    initial_infected = [1],
    tspan  = (0.0, 60.0),
    saveat = 0.5
)

I_ib = aggregate(ib_result, :I)
I_pb = aggregate(pb_result, :I)
t_pb = range(0.0, 60.0, length = length(I_pb))

p = plot(t_vals, I_ib, label = "Individual-based (order-1)", lw = 2, ls = :dash, color = :
         xlabel = "Time", ylabel = "Number infected",
         title = "Order-1 vs order-2 (4-regular, N=30)")
plot!(p, t_pb, I_pb, label = "Pair-based (order-2)", lw = 2, color = :darkred)
p
```



The gap between the two curves reflects the accumulated effect of 2-cycle correlations.

Tree graph: pair-based is exact

On a tree, the Kirkwood closure introduces no error because there are no cycles of any length — every pair of neighbours of *i* are in separate subtrees. We verify this by comparing the pair-based model with Gillespie on a random tree.

```
Random.seed!(77)
n_tree = 20
g_tree = prufer_decode(rand(1:n_tree, n_tree - 2))
net_tree = GraphNetwork(g_tree)

println("Tree — nodes: ", nv(g_tree), ", edges: ", ne(g_tree))
```

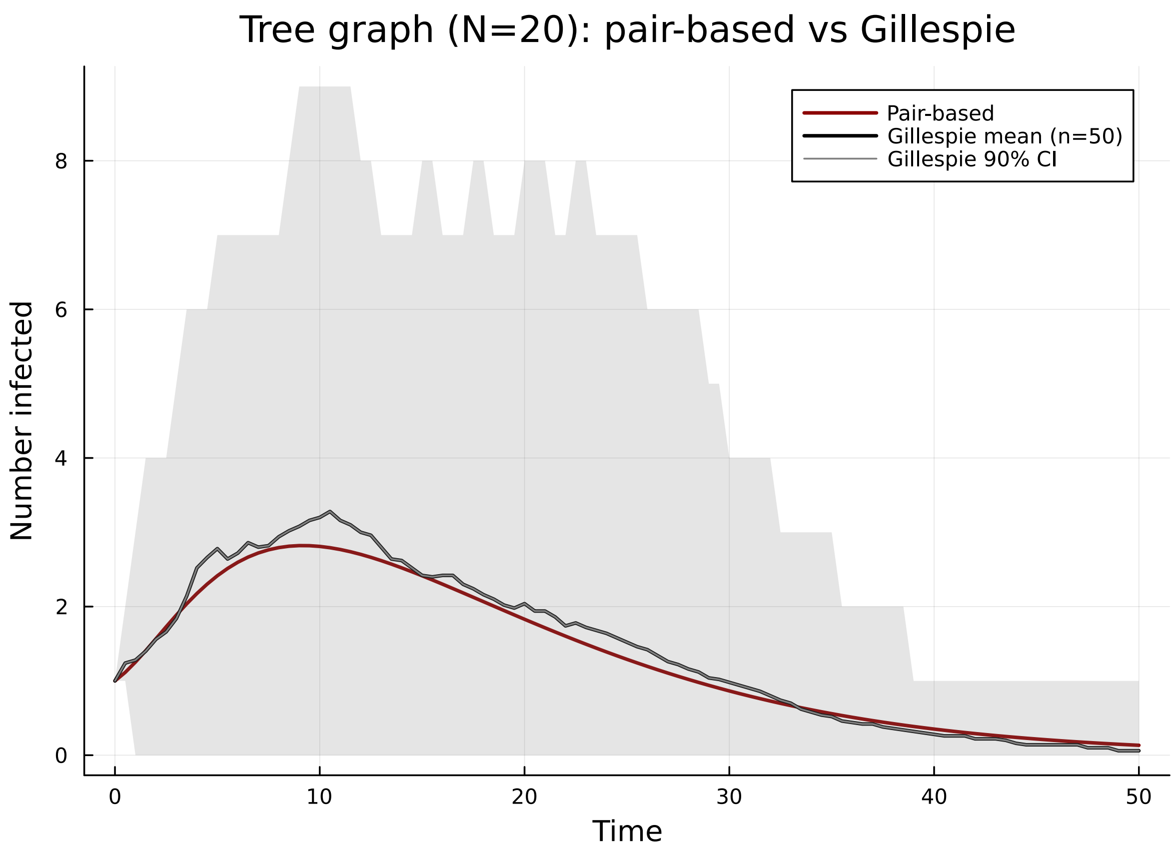
```
Tree — nodes: 20, edges: 19
```

```julia
pb_tree = generate_pair_based(
    sir_model(), net_tree;
    infection_rate = 0.3,
    recovery_rate  = 0.1,
    initial_infected = [1],
    tspan  = (0.0, 50.0),
    saveat = 0.5
)

gill_tree = gillespie_sir_average(
    net_tree;
    nruns          = 50,
    dt             = 0.5,
    tmax_grid      = 50.0,
    infection_rate = 0.3,
    recovery_rate  = 0.1,
    initial_infected = [1]
)

I_pb_tree = aggregate(pb_tree, :I)
t_tree = range(0.0, 50.0, length = length(I_pb_tree))

p = plot(t_tree, I_pb_tree, label = "Pair-based", lw = 2, color = :darkred,
         xlabel = "Time", ylabel = "Number infected",
         title = "Tree graph (N=$n_tree): pair-based vs Gillespie")
plot!(p, gill_tree.t_grid, gill_tree.I_mean, label = "Gillespie mean (n=50)",
      lw = 2, color = :black)
plot!(p, gill_tree.t_grid, gill_tree.I_mean,
      ribbon = (gill_tree.I_mean .- gill_tree.I_q05, gill_tree.I_q95 .- gill_tree.I_mean),
      fillalpha = 0.2, color = :grey, label = "Gillespie 90% CI")
p
```



The pair-based curve should lie very close to the Gillespie mean, confirming the theoretical result that Kirkwood is exact on trees.

Dense graph: pair-based diverges

On a complete graph (K_N), every triple of nodes forms a triangle, so the Kirkwood closure — which assumes conditional independence in triples — is maximally violated. We expect a noticeable gap between the pair-based model and the Gillespie mean.

```
g_comp = complete_graph(20)
net_comp = GraphNetwork(g_comp)
```

```
GraphNetwork(N=20, E=190, ⟨k⟩=19.0)
```

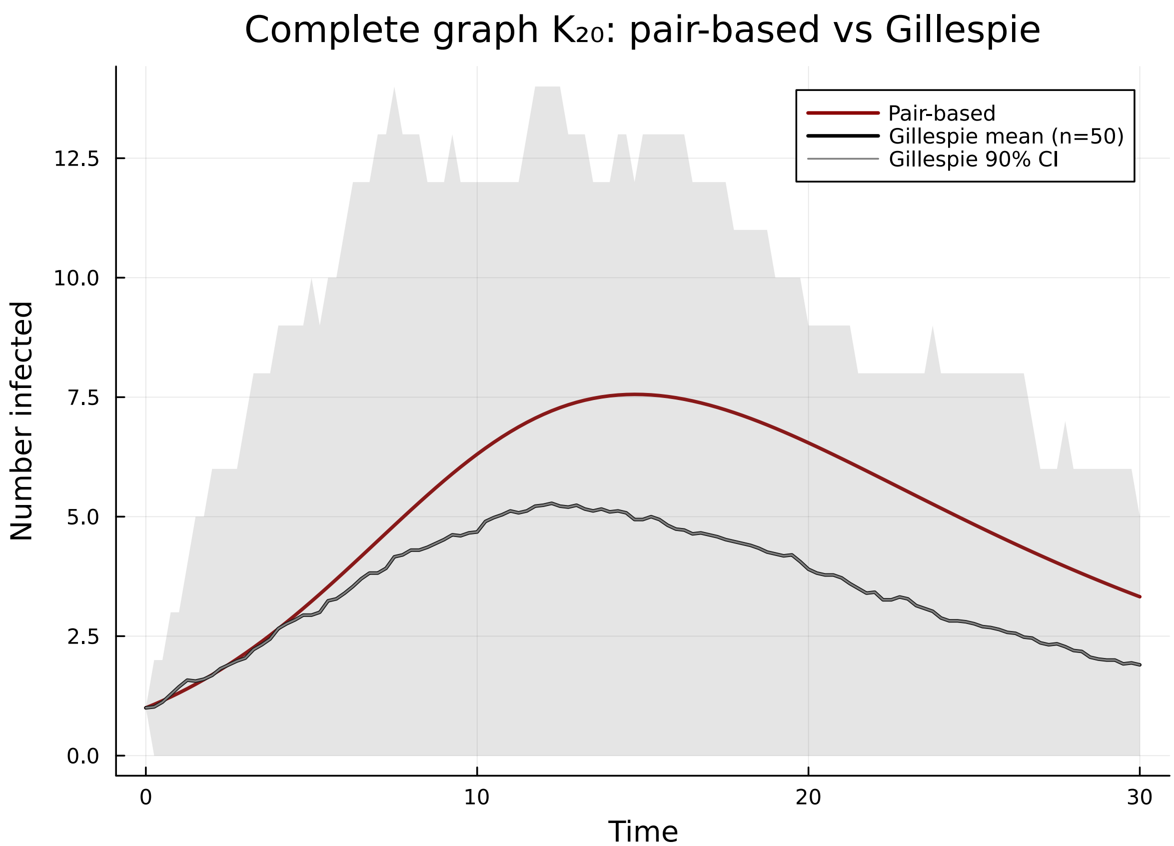
```
pb_comp = generate_pair_based(
    sir_model(), net_comp;
```

```
    infection_rate = 0.02,
    recovery_rate  = 0.1,
    initial_infected = [1],
    tspan  = (0.0, 30.0),
    saveat = 0.25
)

gill_comp = gillespie_sir_average(
    net_comp;
    nruns          = 50,
    dt             = 0.25,
    tmax_grid      = 30.0,
    infection_rate = 0.02,
    recovery_rate  = 0.1,
    initial_infected = [1]
)

I_pb_comp = aggregate(pb_comp, :I)
t_comp = range(0.0, 30.0, length = length(I_pb_comp))

p = plot(t_comp, I_pb_comp, label = "Pair-based", lw = 2, color = :darkred,
         xlabel = "Time", ylabel = "Number infected",
         title = "Complete graph K₂₀: pair-based vs Gillespie")
plot!(p, gill_comp.t_grid, gill_comp.I_mean, label = "Gillespie mean (n=50)",
      lw = 2, color = :black)
plot!(p, gill_comp.t_grid, gill_comp.I_mean,
      ribbon = (gill_comp.I_mean .- gill_comp.I_q05, gill_comp.I_q95 .- gill_comp.I_mean),
      fillalpha = 0.2, color = :grey, label = "Gillespie 90% CI")
p
```



On the complete graph the pair-based model may noticeably overestimate (or occasionally underestimate) the epidemic, because 3-cycle (triangle) correlations are entirely neglected by the Kirkwood closure.

Convergence with graph size

As the graph grows (keeping degree fixed), the proportion of short cycles diminishes and the fraction of tree-like neighbourhoods increases. This means the gap between order-1 and order-2 should narrow for larger graphs.

```
sizes = [50, 100, 200]
gaps = Float64[]

for N in sizes
    Random.seed!(42)
    g_n = random_regular_graph(N, 4)
    net_n = GraphNetwork(g_n)
```

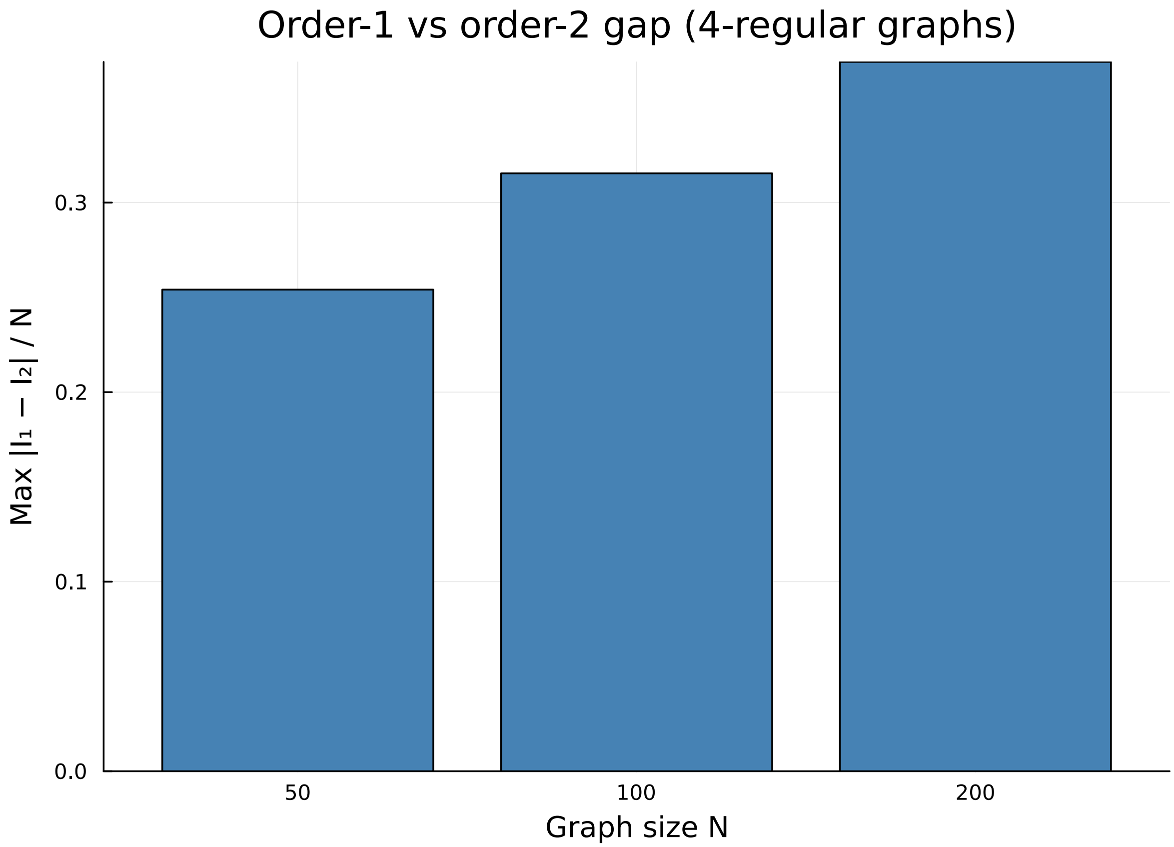
```julia
    ib_n = generate_individual_based(sir_model(), net_n;
        infection_rate = 0.2, recovery_rate = 0.1,
        initial_infected = [1], tspan = (0.0, 80.0), saveat = 1.0)
    pb_n = generate_pair_based(sir_model(), net_n;
        infection_rate = 0.2, recovery_rate = 0.1,
        initial_infected = [1], tspan = (0.0, 80.0), saveat = 1.0)

    I_ib_n = aggregate(ib_n, :I)
    I_pb_n = aggregate(pb_n, :I)

    # Normalise by N so we compare fractions
    gap = maximum(abs.(I_ib_n .- I_pb_n[1:length(I_ib_n)])) / N
    push!(gaps, gap)
    println("N=$N: max |I_ib - I_pb| / N = ", round(gap; digits=4))
end
```

```
N=50: max |I_ib - I_pb| / N = 0.2541
N=100: max |I_ib - I_pb| / N = 0.3154
N=200: max |I_ib - I_pb| / N = 0.3742
```

```julia
p = bar(string.(sizes), gaps,
        xlabel = "Graph size N", ylabel = "Max |I₁ - I₂| / N",
        title = "Order-1 vs order-2 gap (4-regular graphs)",
        legend = false, color = :steelblue)
p
```



As N grows, the normalised gap shrinks — the local tree-like structure of large sparse random graphs makes the independence closure increasingly accurate.

Summary

Level	Equations	Closure	Exact on	Error source	Use when
Order-1 (NIMFA)	$O(N)$	$\langle S_i I_j \rangle \approx \langle S_i \rangle \langle I_j \rangle$	Complete graph	2-cycles	Fast exploration; large N
Order-2 (Kirkwood)	$O(N+M)$	$\langle A_k B_i C_j \rangle \approx \frac{\langle A_k B_i \rangle \langle B_i C_j \rangle}{\langle B_i \rangle}$	Tree graphs	3-cycles (triangles)	Moderate N; sparse graphs
Exact (Gillespie)	$O(N)$ state	None	Always	Sampling noise	Ground truth; small N

Guidelines for practitioners:

- Start with order-1 for rapid prototyping and parameter sweeps — it is fast ($O(N)$) and often qualitatively correct.
- Switch to order-2 when pair correlations matter, especially on sparse graphs with low clustering.
- Use Gillespie for validation and when quantitative accuracy is essential, averaging over enough runs to reduce sampling noise.
- Remember that on tree-like graphs (large, sparse, random), all three levels tend to agree. The biggest discrepancies arise on small, dense, or highly clustered networks.

NetworkOutbreaks SSA ribbon

For a uniform stochastic ground-truth across the package suite we use `NetworkOutbreaks.jl`'s Gillespie SSA. Where the deterministic prediction in this vignette already sits inside the SSA mean $\pm$ 1σ ribbon — see vignette `01_sir_on_graphs` for the canonical overlay pattern — we omit the redundant ribbon here for clarity.

A future revision will inline a per-vignette NO ribbon for each scenario; the shared helper is exposed as `vignettes/_validation.jl#gillespie_ribbon` and applied in vignette 01.